Box Office Gambler

A Monte Carlo Greenlight Simulator

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1. The Problem

A film studio deciding whether to greenlight a movie is making a classic decision under uncertainty. The budget is fixed, but everything else — how big the opening weekend will be, whether audiences show up after week one, how the film performs overseas — is uncertain. Studios deal with this by relying on pattern matching from past experience and gut instinct, which is exactly the kind of heuristic-driven reasoning we studied in the Kahneman and Tversky material on framing effects and anchoring. An executive who just greenlit a $200M hit is going to anchor on that success and overweight the next similar pitch. An executive coming off a flop does the opposite.

I want to build a tool that replaces that heuristic reasoning with a structured Monte Carlo simulation, while keeping the decision process fully transparent — no black boxes, no hidden logic.

2. What the System Does

Box Office Gambler takes in a set of parameters describing a hypothetical movie (production budget, genre, release month, lead actor star power, franchise status, MPAA rating) and runs thousands of simulated financial outcomes. Each trial samples from empirical distributions calibrated to real box office patterns: how opening weekends relate to budgets, how different genres hold up week-over-week, how well certain types of films travel internationally, and so on.

The output is a greenlight decision — one of **GREEN LIGHT**, **PROCEED WITH CAUTION**, or **PASS** — along with the full probability distributions that generated it.

This maps directly onto the course's goals/plans/resources framework. The studio's *goal* is profitability (or more precisely, maximizing the probability of recouping the investment). The *plan* is the configuration of the movie itself — genre, timing, casting tier. The key *resource* is the production budget, which constrains what's possible and defines the threshold for success. The simulation evaluates how likely the plan is to achieve the goal given the resource constraints.

3. How It Explains Decisions

Explanation is the core design concern, and I'm building it into the system at three levels. This connects to what we discussed about explanation quality — specifically, moving from mindless explanations (just restating a conclusion) toward mindful ones (showing the reasoning and the conditions under which the conclusion might change).

Factor attribution. After each simulation run, the system generates a plain-English assessment of each input factor — tagged as positive, negative, or neutral — that explains its specific impact on the outcome. For example: "Horror films have high total multipliers (3.5x budget on average), but are heavily front-loaded — 52% of domestic gross lands in opening weekend, leaving little room for recovery if word-of-mouth is weak." These are direct readouts of the model's actual parameters, not post-hoc rationalizations. The explanation *is* the model.

Probability distributions. Instead of giving a single number ("this movie will gross $300M"), the system shows the full histogram of simulated outcomes with break-even and median lines marked. This is a direct counter to the anchoring bias we covered in the behavioral economics lectures — a point estimate invites anchoring, but a distribution forces the user to confront the spread of possible outcomes. Showing a user that there's a 15% chance of losing everything alongside a 30% chance of a blockbuster is fundamentally different from showing them the average of those two scenarios.

Counterfactual analysis. The system re-runs the simulation across alternative values for genre and release month, holding everything else constant, and displays the comparison. This answers "what would change if I made a different choice?" — which is the structure of a counterfactual explanation. If a user picks September for their action movie and the counterfactual chart shows that June would have added $80M to the median worldwide gross, that's a concrete, actionable piece of reasoning that explains not just what the decision is but why the alternative would be better.

None of these methods require a secondary interpretability model or a surrogate approximation. Monte Carlo simulation is inherently transparent — the inputs are visible, the sampling process is straightforward, and the outputs decompose cleanly into the factors that produced them. This is an advantage over the machine-learning-plus-mimic-model approach listed in the project options: there is no gap between the model and its explanation.

4. Connection to Course Techniques

The project falls under suggested topic #11 (Monte Carlo simulation) and draws on several threads from the course:

Decision strategies. The greenlight decision maps to a decision strategy with explicit criteria (probability of profit above 70% = green light, between 45–70% = caution, below 45% = pass). These thresholds are themselves debatable — a risk-tolerant studio might accept 50% — which opens the door to meta-strategy discussion about how the decision criteria themselves should be chosen.

Behavioral economics. The system is designed to counteract specific biases. Anchoring: distributions instead of point estimates. Framing: the same information presented as both "probability of profit" and "probability of flop" to prevent a one-sided frame from dominating. Representativeness: the counterfactual analysis prevents users from pattern-matching on a single salient feature (e.g., "action movies always do well") by showing the full comparison.

Explanation quality. The factor attributions aim for the mindful end of Langer's spectrum. A mindless explanation would be "the model says pass." A mindful one says "the model says pass because your $250M budget requires roughly $500M worldwide to break even, and the September release window historically depresses opening weekends by 40%, making that target unlikely — here's the probability distribution showing why."

Qualitative reasoning. The factor labels (positive/negative/neutral) and the direction-of-effect descriptions ("this helps," "this hurts," "this is roughly average") are a form of qualitative assessment layered on top of the quantitative simulation. The user gets both — the numbers and the qualitative interpretation of what those numbers mean in context.

5. Implementation Plan

The project is a Python web app built with Streamlit, organized in three modules:

- `data.py` — Empirical priors for the simulation: genre multipliers, seasonal release factors, star power tiers, franchise effects, international performance ratios, and audience ceiling by rating. These are calibrated to aggregate box office data from 2010–2024.
- `engine.py` — The Monte Carlo engine. Accepts a movie description, runs N trials (default 10,000), returns probability distributions and computed metrics. Also contains the factor attribution logic and the greenlight decision function.

- `app.py` — The Streamlit interface. Sidebar for parameter input, main area for the decision banner, distribution histograms (Plotly), percentile tables, factor attribution panel, and counterfactual comparison charts.

Dependencies are minimal: streamlit, numpy, plotly. Everything runs in the Zoo environment with a standard pip install; no additional library approval is needed.

I already have a working prototype that runs locally and produces correct simulation outputs. Remaining work is polishing the counterfactual analysis section and writing the final documentation.